An input-channel audit of RoseTTAFold3: MSA writing, mutation-effect prediction, and conformational-state selection

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Abstract

AlphaFold3-style structure predictors discard the explicit multiple sequence alignment (MSA) representation after a compact module within each recycle, leaving the pair representation to carry evolutionary information. Here we audit this input channel in RoseTTAFold3 across representation geometry, deep mutational scanning (DMS), and conformational-state selection. Across three proteins, MSA input reduced effective rank by 16–42% while preserving mutant-conditioned directions and improved cross-position DMS prediction. De-biased gains were +0.14–0.15 Spearman ρ on RBD and +0.14–0.18 on CASP3; contiguous-block cross-validation yielded gains of +0.256 and +0.264, respectively. A single MSA recycle installed approximately 75% of the geometric and 85% of the functional effect, with only about 25% of the geometric effect erased by nine subsequent MSA-free recycles. On RBD, conservation contributed approximately two-thirds and covariation one-third of the alignment-content gain, which approached saturation near eight rows. Distogram analysis of 13 two-state entries and explicit sampling of 480 structures showed that MSA effects on conformational selection were target-dependent and consistent with family consensus. Structural templates constituted a distinct prior: they impaired RBD mutation-effect prediction and strongly steered sampled conformations, but no universal template–MSA hierarchy was observed. Native MSAs preserved or restored the native fold in GA88 and produced a mixed ensemble in GB88 when opposed by an alternative-fold template. Several template-conditioned distogram preferences disagreed with sampled coordinates. These results establish a multi-level account of MSA and template priors in RoseTTAFold3 and show that conformational audits should include explicit coordinate generation.

Author Summary

Proteins can adopt different shapes to perform their functions, and computational models are increasingly used to predict these shapes and the effects of mutations. These models can use information from many related protein sequences, but it has been unclear how this information is retained inside newer model architectures. We investigated RoseTTAFold3 and found that evolutionary information leaves a lasting imprint after a single processing step. This imprint improves the prediction of mutation effects and can influence which of several possible protein shapes the model selects. Supplying an example structure, known as a structural template, affects the model in a different way: it can strongly guide the predicted shape but reduce sensitivity to mutations. The outcome of combining evolutionary information and structural templates depends on the protein. We also found that an internal distance-based score can disagree with the three-dimensional structures that the model actually generates. Our findings provide a unified way to examine how different inputs guide structure prediction and highlight the importance of checking generated structures directly.

1 Introduction

Deep learning models for biomolecular structure prediction rely on three primary input channels: the target amino acid sequence, homologous multiple sequence alignments (MSAs) [1, 2], and homologous or experimental structural templates [3, 4]. In the canonical AlphaFold2 (AF2) architecture, the Evoformer maintains a full, explicit MSA representation alongside the residue-pair representation across all 48 blocks, continuously exchanging information between sequence alignments and pairwise geometric features [3]. However, recent architectural generations—including AlphaFold3 (AF3) [4], RoseTTAFold3 (RF3) [5], and related Pairformer designs [6]—reduce the computational footprint of alignment processing. In an AF3-style trunk, the MSA is processed only within a compact four-block module at the beginning of each recycle via outer-product mean and transition operations;

thereafter, the explicit MSA representation is discarded, and downstream reasoning within the Pairformer trunk and diffusion structure module relies on the residue-pair representation.

This architectural design raises several mechanistic questions. When a transient MSA input is projected into the pair representation and subsequently discarded, what geometric and biological information is written? Does the pair representation require repeated MSA reinforcement across all recycles, or does the initial recycle establish a persistent state that subsequent MSA-free recycles only partially modify? Furthermore, how does this evolutionary write simultaneously affect two distinct downstream tasks: predicting the functional consequences of single-point mutations—which requires sensitivity to local sequence perturbations [7–9]—and selecting among alternative global folds or conformational states in polymorphic proteins [10–13]?

A parallel set of questions concerns how the MSA channel compares to structural templates and pretrained protein language model (pLM) embeddings [14, 15]. In structure prediction and ensemble generation workflows, MSAs and templates are often used as complementary restraints to steer conformational sampling [16–20]. Yet from an information-theoretic perspective, an alignment of diverged homologs encodes family-wide sequence statistics (encompassing both site-wise conservation and pairwise covariation), whereas a single structural template provides fixed atomic coordinates corresponding to one specific state. Whether a template acts as an equivalent evolutionary prior or as a mutation-suppressing structural constraint within an AF3-style pair representation—and how conflicts between templates and MSAs resolve across different targets—requires systematic evaluation across representation, mutational, and conformational levels.

Here, we perform a multi-level input-channel audit of RoseTTAFold3 (RF3) [5]. Using a single fixed checkpoint and inference configuration, we trace how input channels modify the trunk pair representation ($\mathbf{Z}_{\mathrm{II}}$) and govern downstream readouts. Across three deep mutational scanning (DMS) benchmarks—SARS-CoV-2 receptor-binding domain (RBD) [7, 21], human executioner caspase-3 (CASP3) [22], and bacteriophage λ repressor CI [9]—alongside a 13-entry two-state conformational panel [23–28] and 480 sampled 3D structures, our experiments support four main conclusions:

1. **Persistent family-consensus writing:** The MSA leaves a moderate, non-erasing compression in the pair representation that is largely installed in a single recycle ($K = 1$) and persists across subsequent MSA-free recycles.
2. **Evolutionary content over mere compression:** This write improves cross-position DMS prediction through conservation and covariation content rather than rank compression per se, holding under de-biased standardization and contiguous-block cross-validation.
3. **Target-dependent conformational selection:** The same family-consensus prior steers multi-state proteins toward or away from crystallized states depending on family evolutionary context, with major fold-switcher directions verified by explicit 3D coordinate sampling.
4. **Distinct template priors and readout decoupling:** Structural templates act as a distinct prior that impairs DMS mutation sensitivity on RBD and exhibits target-dependent conflict resolution against MSAs in 3D coordinate sampling, while showing that distogram log-likelihood readouts can disagree with sampled 3D coordinates in template-conditioned arms.

2 Results

2.1 The MSA leaves a persistent, one-shot family-consensus write in the pair representation

We first examined how supplying an MSA alters the geometry of single-mutant pair representations ($\mathbf{Z}_{\mathrm{II}}$) and whether this geometric reorganization improves or degrades mutation-effect prediction. Across all three DMS benchmarks, introducing the MSA induced a moderate dimensionality compression without collapsing the representation ensemble (Table 1; S1 Fig). Specifically, the effective rank (`eff90`, the number of principal components explaining 90% of centered mutant ensemble variance) decreased by 26% in SARS-CoV-2 RBD (231 → 171), 42% in homodimeric CASP3 (333 → 193), and 16% in λ repressor CI (61 → 51). Linear centered kernel alignment (CKA) [29] between the no-MSA and with-MSA ensembles ranged from 0.527 (RBD) and 0.553 (CASP3) to 0.917 (CI), while median directional cosine similarity of individual mutation difference vectors (`dCos`) ranged from 0.362 to 0.844. In full-set RBD, where per-mutant norm ratios were evaluated, the median Euclidean norm of mutant-minus-wild-type perturbation vectors increased rather than shrank (Δ norm ratio = 1.21). Thus, the MSA did not act as a sequence-erasing filter that projected point mutants onto an identical wild-type representation; instead, it re-oriented mutant perturbations within a more compact subspace.

Functionally, this geometric reorganization improved cross-position DMS prediction across all three targets (Table 1). Under the canonical full-set evaluation, MSA addition increased Spearman ρ by +0.179 (and +0.204 in

Table 1: Geometric reorganization and cross-position DMS predictive gain induced by MSA input across three benchmark proteins.

Protein	`eff90` (no → MSA)	Rel. Δ`eff90`	Linear CKA	`dCos`	Canonical $\Delta\rho$ (MSA − no)	De-biased $\Delta\rho$
SARS-CoV-2 RBD	231 → 171	−26%	0.527	0.474	+0.179; rep. +0.204 (5/5 splits)	+0.14 to +0.15
Human CASP3	333 → 193	−42%	0.553	0.362	+0.162 (5/5 splits)	+0.14 to +0.18
λ Repressor CI	61 → 51	−16%	0.917	0.844	+0.075 (4/5 splits)	Not fully de-biased

an independent full-set replicate, positive in 5/5 splits) on RBD, +0.162 (5/5 splits) on CASP3, and +0.075 (4/5 splits) on the smaller CI dataset ($N = 351$, best-validation protocol).

Because AF3-style trunks re-inject the MSA at the start of each recycle, we performed recycle-gating experiments on the 500-mutant stride-8 RBD subset (Protocol E7; Table 2B; S2 Fig) in which the 15-row MSA was provided for only the first $K \in \{1, 2, 3\}$ recycles, followed by $10 - K$ MSA-free recycles. Exposing the trunk to the MSA for a single recycle ($K = 1$) compressed `eff90` from 181 to 150—capturing 74% (approximately 75%) of the total geometric compression achieved by 10 full MSA recycles (`eff90` = 139)—and raised DMS Spearman ρ from 0.414 to 0.605, capturing 85% of the full 10-recycle functional gain ($\rho = 0.638$). Extending exposure to $K = 2$ and $K = 3$ recycles captured 86–88% of the geometric shift and 87–96% of the functional gain (with minor non-monotonic variation between $K = 2$ and $K = 3$). Nine subsequent MSA-free recycles following $K = 1$ washed out only about 25% of the initial geometric compression. Within the 10-recycle inference regime, MSA writing into the pair representation is therefore predominantly a one-shot event that persists across subsequent recycles.

Table 2: Dissection of MSA dose, biological content (Protocol E1, Panel A), and recycle-gating dynamics (Protocol E7, Panel B) on the RBD 500-mutant stride-8 subset.

Arm / Exposure	Protocol Description & Perturbation	`eff90`	Spearman ρ	Gain / Share
Panel A: Protocol E1 Dose × Content Decomposition *(Gain referenced to `depth01` zero-content baseline)*				
`noMSA`	Sequence track only (MSA pathway bypassed)	181	0.414	−0.076
`depth01`	Single query row via MSA pathway (zero-content baseline)	165	0.490	0.000
`random`	Shared alignment context; synthetic random sequences	149	0.479	−0.011
`rowshuffle`	Shared alignment context; per-row shuffled residues	155	0.477	−0.013
`consensus`	Identical copies of the MSA consensus sequence (profile mode)	156	0.501	+0.011
`colshuffle`	Per-column residue frequencies preserved; covariation destroyed	150	0.596	+0.106
`depth04`	4 real homologous sarbecovirus rows	157	0.536	+0.046
`depth08`	8 real homologous sarbecovirus rows	138	0.626	+0.136
`full15`	All 15 real homologous sarbecovirus rows	139	0.638	+0.148
Panel B: Protocol E7 Recycle Gating *(Geometric / functional share of 10-recycle `full15` effect vs. `noMSA`)*				
$K = 1$ recycle	1 MSA recycle followed by 9 MSA-free recycles	150	0.605	74% geom. / 85% func.
$K = 2$ recycles	2 MSA recycles followed by 8 MSA-free recycles	145	0.628	86% geom. / 96% func.
$K = 3$ recycles	3 MSA recycles followed by 7 MSA-free recycles	144	0.609	88% geom. / 87% func.

2.2 DMS gains arise from conservation and covariation rather than rank compression or split leakage

To test whether the observed DMS gains were inflated by validation protocol optimism or sequence-position proximity between training and test sets, we evaluated de-biasing schemes and contiguous-block cross-validation (Protocol E427; S1 Table). First, separating epoch selection from test evaluation and enforcing strict train-only standardization measured two sources of upward bias in canonical best-validation numbers: single-validation best-epoch selection contributed +0.04 to +0.09 across cells (up to +0.094 on the RBD MSA-only probe across 5/5

splits), while full-sample channel standardization shifted RBD MSA-only ρ from 0.589 to 0.644 (+0.055 across 15/15 paired runs) compared to +0.022 for no-MSA. Removing both biases preserved a consistent MSA advantage: on RBD, held-out test evaluation at the validation-selected epoch (+0.141), best-test evaluation (+0.143), and strict train-only standardization (+0.146) yielded de-biased gains of +0.14 to +0.15. On CASP3, dual-validation de-biased estimates yielded +0.142 to +0.183. Comparing RBD and CASP3 indicated that validation-set asymmetry is not determined simply by rank compression: CASP3 underwent stronger rank compression than RBD (−42% vs. −26%) without exhibiting the same standardization shift, and the two proteins showed opposite relationships between stable and noisy single-stream probes.

Second, replacing random position holdout with 5-fold contiguous sequence-block cross-validation (Protocol E427; S1 Table) yielded net MSA gains of +0.256 on full-set RBD (5/5 folds: +0.200, +0.328, +0.312, +0.214, +0.226) and +0.264 on CASP3 (5/5 folds: +0.330, +0.141, +0.267, +0.351, +0.231), exceeding the corresponding random-position holdout gains (+0.204 and +0.162). Random position holdout therefore did not manufacture the MSA advantage; rather, random splits modestly aided the no-MSA baseline via interpolation from adjacent sequence positions. Evaluating lightweight probes in E427 showed that a per-residue multilayer perceptron (MLP) preserved the MSA gain across all three proteins (+0.157 on RBD, +0.160 on CASP3, +0.089 on CI; all 5/5 splits). A flattened linear probe retained the gain on CASP3 (+0.163, 5/5 splits) and directionally on CI (+0.085, 3/5 splits), but dropped to +0.030 on RBD (3/5 splits, with both arms near $\rho \approx 0.30$), defining a readout boundary where RBD's MSA write requires at least per-residue nonlinear decoding.

To determine whether rank compression itself drove functional gains, we examined the 9-arm dose × content decomposition on the 500-mutant RBD subset (Protocol E1; Table 2A; S2 Fig). Passing the single wild-type sequence through the MSA featurization pathway (`depth01`) reduced `eff90` from 181 (`noMSA`) to 165 and raised ρ from 0.414 to 0.490 (+0.076), identifying a featurization-path effect and establishing `depth01` ($\rho = 0.490$) as the zero-biological-content baseline. Relative to `depth01`, alignment effects separated into three components:

1. **Shared-context compression tax:** Supplying `random` or `rowshuffle` alignments compressed `eff90` to 149–155 without improving DMS prediction (−0.011 and −0.013 vs. `depth01`), showing that rank reduction alone does not confer functional gain.
2. **Conservation (≈ 2/3 of biological gain on RBD):** Preserving per-column amino acid frequencies while destroying inter-column covariation (`colshuffle`) yielded $\rho = 0.596$ (+0.106 over `depth01`). Because a modal consensus sequence (`consensus`) gave +0.011, the conservation-specific step from `consensus` to `colshuffle` contributed +0.095, or roughly two-thirds of the +0.148 biological gain. Although `consensus` and `colshuffle` had similar effective ranks (156 vs. 150), their mutant directional cosines differed substantially (`dCos` = 0.815 vs. 0.548), indicating that directional reorganization rather than rank explains their functional difference. Notably, $K = 1$ recycle and `colshuffle` both yielded `eff90` = 150, providing a geometric triangulation without implying identical information content.
3. **Covariation (≈ 1/3 of biological gain) and saturation:** Restoring intact homologous rows (`colshuffle` → `full15`) added +0.042 ($\rho = 0.596 \rightarrow 0.638$). Across real row depths, gains rose from +0.046 (`depth04`) to +0.136 (`depth08`) and +0.148 (`full15`), approaching saturation near 8 rows. Across RBD (15 rows, CKA 0.527), CASP3 (69 rows, CKA 0.553), and CI (512 rows, CKA 0.917), DMS gains tracked the magnitude of representation rewriting, consistent with family coherence rather than raw alignment depth being the operative variable in these systems.

2.3 Family consensus shapes conformational-state selection in fold-switching proteins

We next investigated how the MSA write affects conformational-state selection across 13 two-state protein entries using range-matched distogram negative log-likelihood (NLL) preferences (pref = $\mathrm{NLL}_B - \mathrm{NLL}_A$; Fig 1A and S2 Table). Across the 13 entries, 8 targets—including four conformational switches (β_2AR, Abl1, EGFR, LacY), the 2-row selecase alignment, and three X-ray/NMR static controls (ubiquitin, GB1, thioredoxin)—showed MSA shifts within the empirical repeat-forward noise band (±0.14 to ±0.27). Large MSA shifts occurred in the fold-switching systems:

- **Away from the crystallized state (2 targets):** In KaiB (pref : +0.72 → −1.57, shift = −2.29) and RfaH (+0.10 → −2.89, shift = −2.99), adding the MSA shifted distogram preference away from the crystallized state (State A) toward the solution NMR fold-switched state (State B).
- **Toward the crystallized state (2 targets):** In Mad2 (−0.05 → −0.96, shift = −0.91, where the crystallized state corresponds to closed State B) and XCL1 (−0.78 → +2.09, shift = +2.87, toward State A), the MSA shifted preference toward the crystallized state.
- **Own-fold sharpening in a designed pair (1 entry):** In GA88/GB88, the main range-matched flow on GA88 showed sharpening toward its 3α fold (+0.52 → +3.07, shift = +2.56). Under the separate E49b

residue-universe scoring flow, GA88 shifted from +0.38 → +2.74 (3α preference) and GB88 shifted from −0.23 → −2.66 (α/β preference).

This bidirectional pattern (2 toward crystal, 2 away, 1 designed-pair sharpening, 8 null) is consistent with the MSA acting as a family-consensus selector rather than a uniform crystal-state bias: in KaiB and RfaH, where the alternative fold-switched form is more prevalent across homologs, the MSA pulls away from the crystallized state, whereas in Mad2 and XCL1 the inferred family consensus aligns with the crystallized state.

To test whether these distogram shifts correspond to actual 3D structures, we sampled 480 structures across 30 condition cells (Protocol E428: 6 target entries × 5 arms × 16 structures/cell, 0 failures; Table 3; Fig 1B and S3 Table). In the MSA-only arms examined by coordinate sampling, sampled Cα structures ($\Delta\text{RMSD} = \text{RMSD}_B - \text{RMSD}_A$) confirmed the primary MSA directions across all four natural fold-switchers: KaiB shifted from 100% closer to A (`nomsa` ΔRMSD = +4.174 Å) to 0% closer to A (`msa` −4.698 Å); RfaH shifted from 56.2% closer to A (+0.185 Å) to 0% closer to A (−9.791 Å); Mad2 shifted from 87.5% closer to A (+0.767 Å) to 0% closer to A (−4.142 Å); and XCL1 shifted from 0% closer to A (−4.000 Å) to 100% closer to A (+6.751 Å). In the designed pair, coordinate sampling showed that sequence alone (`nomsa`) placed GB88 in its native B fold (0% closer to A, −3.058 Å) but did not stably select GA88's native A fold (only 6.2% closer to A, −2.179 Å). Supplying each protein's own MSA resulted in 100% native-state selection for both GA88 (+6.682 Å, 100% A) and GB88 (−12.387 Å, 0% A / 100% B).

Table 3: Summary of explicit 3D coordinate sampling across six fold-switching systems (Protocol E428: 30 cells × 16 structures = 480 structures) compared against distogram NLL readouts. Positive $\Delta\text{RMSD} = \text{RMSD}_B - \text{RMSD}_A$ indicates 3D coordinates closer to State A (for GA88/GB88, A is the GA88 3α fold and B is the GB88 α/β fold). Full per-cell RMSD A and RMSD B values are provided in S3 Table.

System (Metric)	`nomsa`	`msa`	`tplA/0wn`	`tplB/0th`	`msa+tpl`	Key Coordinate vs. Distogram Audit Outcome
KaiB (ΔRMSD, Å)	+4.174	−4.698	+6.752	−5.871	+6.349	3D & distogram concordant; `msa_tplA` follows `tplA`
Closer to A (%)	100.0%	0.0%	100.0%	0.0%	100.0%	(`pref`: +0.72, −1.57, +3.59, −3.00, +3.36)
RfaH (ΔRMSD, Å)	+0.185	−9.791	+12.879	−6.369	+7.598	`tplB` strongly steers 3D coords despite +0.30 distogram
Closer to A (%)	56.2%	0.0%	100.0%	0.0%	100.0%	(`pref`: +0.10, −2.89, +1.49, +0.30, +1.45)
Mad2 (ΔRMSD, Å)	+0.767	−4.142	+8.046	−9.643	+7.730	Resolves `tplA` distogram anomaly (−0.17): 16/16 at A
Closer to A (%)	87.5%	0.0%	100.0%	0.0%	100.0%	(`pref`: −0.05, −0.96, −0.17, −1.87, −0.17)
XCL1 (ΔRMSD, Å)	−4.000	+6.751	+4.919	−3.453	+4.526	`tplA` steers 16/16 to A (inverting −1.49 distogram);
Closer to A (%)	0.0%	100.0%	100.0%	6.2%	93.8%	`msa` & `tplA` co-directional in 3D despite opposite distograms
GA88 (ΔRMSD, Å)	−2.179	+6.682	+6.923	−12.378	+5.593	`msa_tplOther` preserves native A (16/16), reversing
Closer to A (%)	6.2%	100.0%	100.0%	0.0%	100.0%	distogram −4.24 (`pref`: +0.38, +2.74, +4.88, −4.30, −4.24)
GB88 (ΔRMSD, Å)	−3.058	−12.387	−12.761	+8.871	−0.796	`msa_tplOther` yields 50/50 split, whereas distogram
Closer to A (%)	0.0%	0.0%	0.0%	100.0%	50.0%	favored A at +4.58 (`pref`: −0.23, −2.66, −4.34, +4.75, +4.58)

2.4 Structural templates are a distinct, target-dependent prior and expose readout decoupling

We next compared structural templates against MSAs in both conformational selection (Table 3; Fig 2A and 2B; S2 and S3 Tables) and RBD mutation-effect prediction (Table 4; Fig 2C and 2D).

In the conformational panel, comparing distogram readouts against 3D sampled coordinates across the template-conditioned arms revealed substantial readout decoupling (Table 3; Fig 2A and 2B):

1. **Resolution of apparent natural template anomalies:** On the distogram readout, supplying Template A (`tplA`) yielded negative preferences in Mad2 (−0.17) and XCL1 (−1.49), and supplying `tplB` in RfaH yielded a near-null value (+0.30). In explicit 3D coordinate sampling, however, `tplA` steered 100% (16/16) of structures to State A in both Mad2 ($\Delta\text{RMSD} = +8.046$ Å, $\text{RMSD}_A = 3.140$ Å) and XCL1 ($\Delta\text{RMSD} = +4.919$ Å, $\text{RMSD}_A = 8.023$ Å), and `tplB` steered 100% (16/16) of RfaH structures to State B ($\Delta\text{RMSD} = -6.369$ Å). Thus, the negative Mad2 and XCL1 `tplA` distogram values were readout artifacts rather than template failures, and in XCL1 the `msa` and `tplA` inputs actually favored the same state (State A) in 3D space.
2. **Target-dependent conflict resolution between MSA and template:** In the three natural fold-switchers where MSA and `tplA` opposed each other in 3D coordinates (KaiB, RfaH, and Mad2), the combined `msa_tplA` arm followed `tplA` (100% closer to State A in all three systems). In the designed pair GA88/GB88, supplying

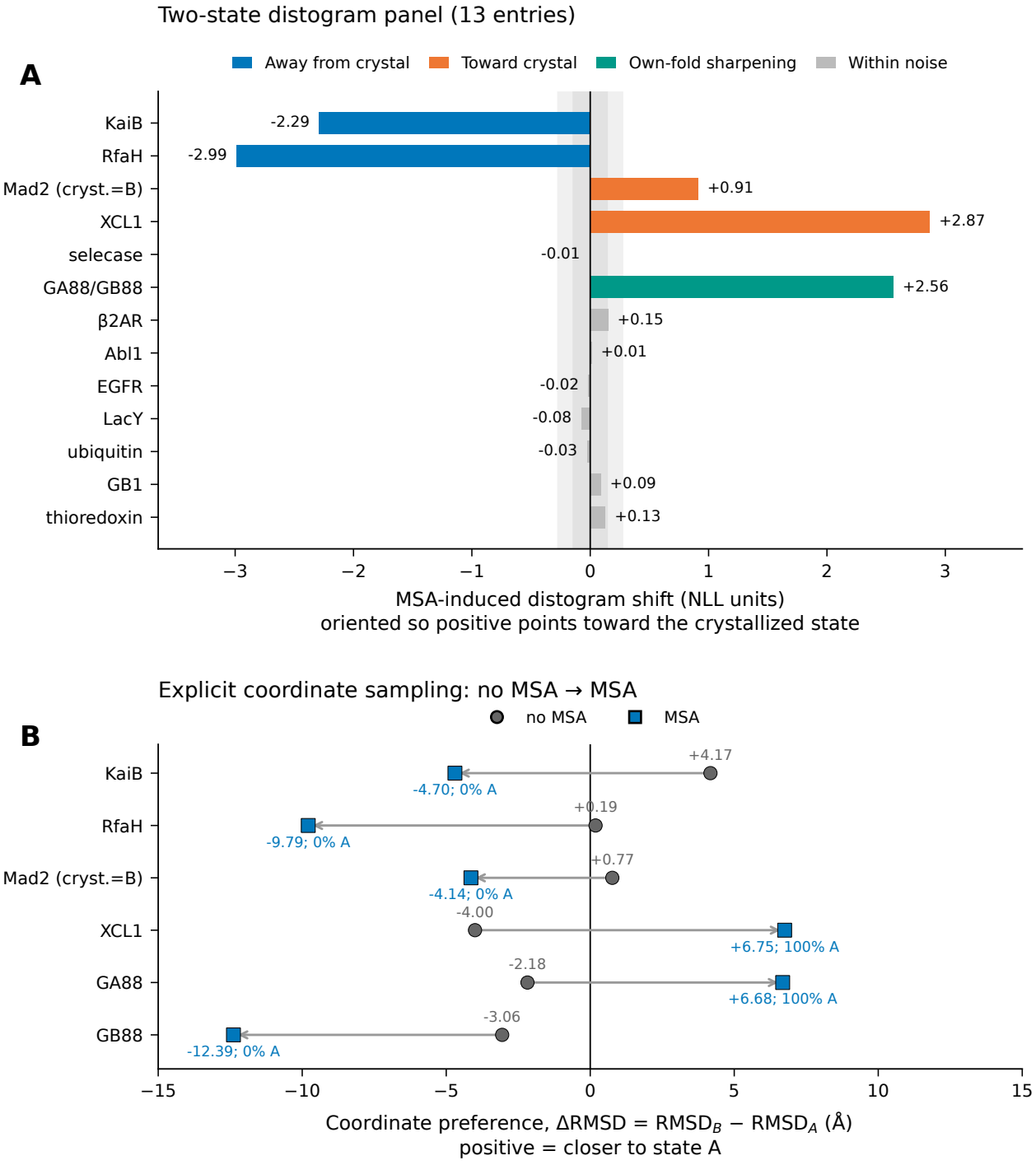


Fig 1. MSA-driven conformational preference shifts across 13 two-state entries and explicit 3D coordinate verification. (A) Range-matched distogram preference shifts ($\Delta\text{pref}_{\text{MSA}}$, in NLL units, oriented so positive values point toward the crystallized state; for Mad2, where the crystallized closed state is State B, raw shift is −0.91 and crystal-oriented shift is +0.91) across the 13-entry two-state panel, categorized into shifts away from crystal (blue: KaiB, RfaH), toward crystal (orange: Mad2, XCL1), own-fold sharpening (teal: GA88/GB88), and shifts within the empirical repeat-forward noise band (±0.14 to ±0.27, shaded gray). (B) Explicit 3D coordinate sampling preferences ($\Delta\text{RMSD} = \text{RMSD}_B - \text{RMSD}_A$ in Å, where positive values indicate structures physically closer to State A; $n = 16$ sampled structures per cell, Protocol E428) comparing sequence-only (`no MSA`, gray circles) to alignment-conditioned (`MSA`, blue squares) generation across KaiB, RfaH, Mad2, XCL1, GA88, and GB88. Labels report mean ΔRMSD and the percentage of sampled structures closer to State A.

the opposite-fold template alone (`tplOther`) flipped 100% (16/16) of sampled structures to the opposite fold (ΔRMSD = −12.378 Å for GA88; +8.871 Å for GB88). When paired with each sequence's own MSA (`msa_tplOther`), distogram scores remained near the opposite-fold template (−4.24 for GA88; +4.58 for GB88), but explicit 3D coordinates showed that GA88 retained its native 3α fold in 100% (16/16) of structures (ΔRMSD = +5.593 Å, RMSD$_A$ = 6.008 Å) while GB88 shifted to a 50.0% (8/16) mixed ensemble (ΔRMSD = −0.796 Å). Consequently, no single cross-target input hierarchy holds across all tested systems.

Table 4: Comparison of MSA and structural template priors on SARS-CoV-2 RBD representation geometry and DMS prediction (500-mutant E49-RBD subset and E427 contiguous block-CV verification).

Input Arm	`eff90`	CKA vs. no	Δ Dir.	Δ Norm	Subset ρ	E427 Contiguous Block-CV Verification ($\Delta\rho$)
`no MSA`	181	1.000	1.00	1.00	0.415	Baseline reference
`MSA (15 rows)`	139	0.568	0.48	1.21	0.638	MSA − no = +0.259 (5/5 folds: +0.121 to +0.373)
`template (6M0J)`	160	0.443	0.26	1.51	0.366	tpl − no = −0.129 (4/5 folds: −0.250 to −0.101, 5th −0.004)
`MSA + template`	159	0.448	0.26	1.52	0.370	MSA+tpl − tpl = +0.003 (null); MSA+tpl − no = −0.126

On the 500-mutant E49-RBD subset, comparing 'no MSA', 'MSA 15 rows', wild-type crystal 'template 6M0J' [21], and 'MSA + template' showed that templates and MSAs had opposite functional effects on DMS prediction (Table 4; Fig 2C and 2D). Supplying template '6M0J' lowered CKA against no-MSA to 0.443 (CKA(template, MSA) = 0.482), reduced mutant directional consistency (Δ direction = 0.26 vs. 0.48 for MSA), increased Δ norm ratio to 1.51, and reduced DMS Spearman ρ from 0.415 to 0.366 ($\Delta\rho$ = −0.049 vs. +0.223 for MSA). Combining 'MSA + template' yielded representation geometry (`eff90` = 159, CKA = 0.448, Δ direction = 0.26, Δ norm = 1.52) and DMS correlation (ρ = 0.370, $\Delta\rho$ = −0.045 vs. no-MSA) close to template alone. Re-evaluating the same four arms under 5-fold contiguous-block cross-validation with the canonical `SinglePredictorV2` head (Protocol E427; Table 4) yielded +0.259 for 'MSA - no' (5/5 folds: +0.121, +0.373, +0.321, +0.223, +0.258), −0.129 for 'template - no' (4/5 folds: −0.250, −0.152, −0.101, −0.138, fifth fold −0.004), +0.003 for 'MSA+template - template' (folds +0.025, −0.042, +0.001, +0.014, +0.016), and −0.126 for 'MSA+template - no'. These observations are consistent with a mutation-erasing template prior on RBD that dominates the combined representation and suppresses the DMS gain from the MSA.

2.5 Supporting information accounting and homodimer interface controls

Four supporting analyses further bounded the information content and behavior of the MSA write (full details in S4 Table and S1 Text):

- **Three-stream marginal accounting ($\mathbf{Z}_{\text{no}}$, $\mathbf{Z}_{\text{msa}}$, `FAESM`):** In cross-attention fusion with ESM-2 650M embeddings (`FAESM`) [14], adding $\mathbf{Z}_{\text{no}}$ to $\mathbf{Z}_{\text{msa}}$+`FAESM` produced no detectable additional marginal contribution under the evaluated probe and protocol (+0.002 on RBD, −0.001 on CASP3, +0.008 on CI). Adding `FAESM` to $\mathbf{Z}_{\text{msa}}$ gave −0.005 on RBD (compared to +0.075 on $\mathbf{Z}_{\text{no}}$), +0.051 (5/5 splits) on CASP3, and an unresolved −0.039 on CI driven by a single outlier split (S4 Table).
- **Gradient dominance vs. paired noise floor (Protocol E429):** Late-training encoder gradients favored `FAESM` over pair streams across all three proteins ($f/z \approx 6.5$, $f/m \approx 2.7$ on RBD; $f/m \approx 3.5$–3.8 on CASP3/CI). Scaling the RBD `FAESM` encoder learning rate by $\lambda = 0.25$ yielded a split-paired gain of $+0.0143 \pm 0.0032$ paired SE ($t = 4.48$, 5/5 splits: +0.0155, +0.0252, +0.0091, +0.0146, +0.0069), with 75% Gaussian input noise offering directional support (0.649 → 0.662, $+0.0128 \pm 0.0110$ SE, 3/5 splits). Protocol E429 (16 identical runs across two processes on RBD `x_msa_faesm`) measured a within-process SD of 0.0016–0.0027 (15/16 runs in 0.6683–0.6732, full range 0.012 with one outlier at 0.6608), showing that while ±0.02 applies to unpaired cell comparisons, the paired learning-rate effect exceeds paired training noise.
- **Partial reconstruction of $\mathbf{Z}_{\text{msa}}$ from $\mathbf{Z}_{\text{no}}$ + `FAESM` (Protocol D1):** Across linear, MLP-256/512, and 1–8-layer transformer reconstructors on RBD, validation CKA plateaued near 0.80–0.83 from MLP-512 onward (linear 0.74; coordinate R^2 rising from 0.25 to 0.36; 8-layer train–validation CKA gap +0.09, suggesting substantial generalization rather than rote memorization; one-hot amino-acid control CKA ≈ 0.50). Separate encoders trained with a CKA objective at depths 2/4/6/8 ordered as both (0.811) > `FAESM`-only (0.75–0.76) > $\mathbf{Z}_{\text{no}}$-only (0.716–0.717). Reconstructed $\hat{\mathbf{Z}}_{\text{msa}}$ achieved ρ = 0.575 vs. 0.446 ($\mathbf{Z}_{\text{no}}$) and 0.663 (real $\mathbf{Z}_{\text{msa}}$), recovering 59.4% (≈ 60%) of the functional advantage while leaving approximately 40% of the MSA-specific functional advantage unrecovered at these model scales.

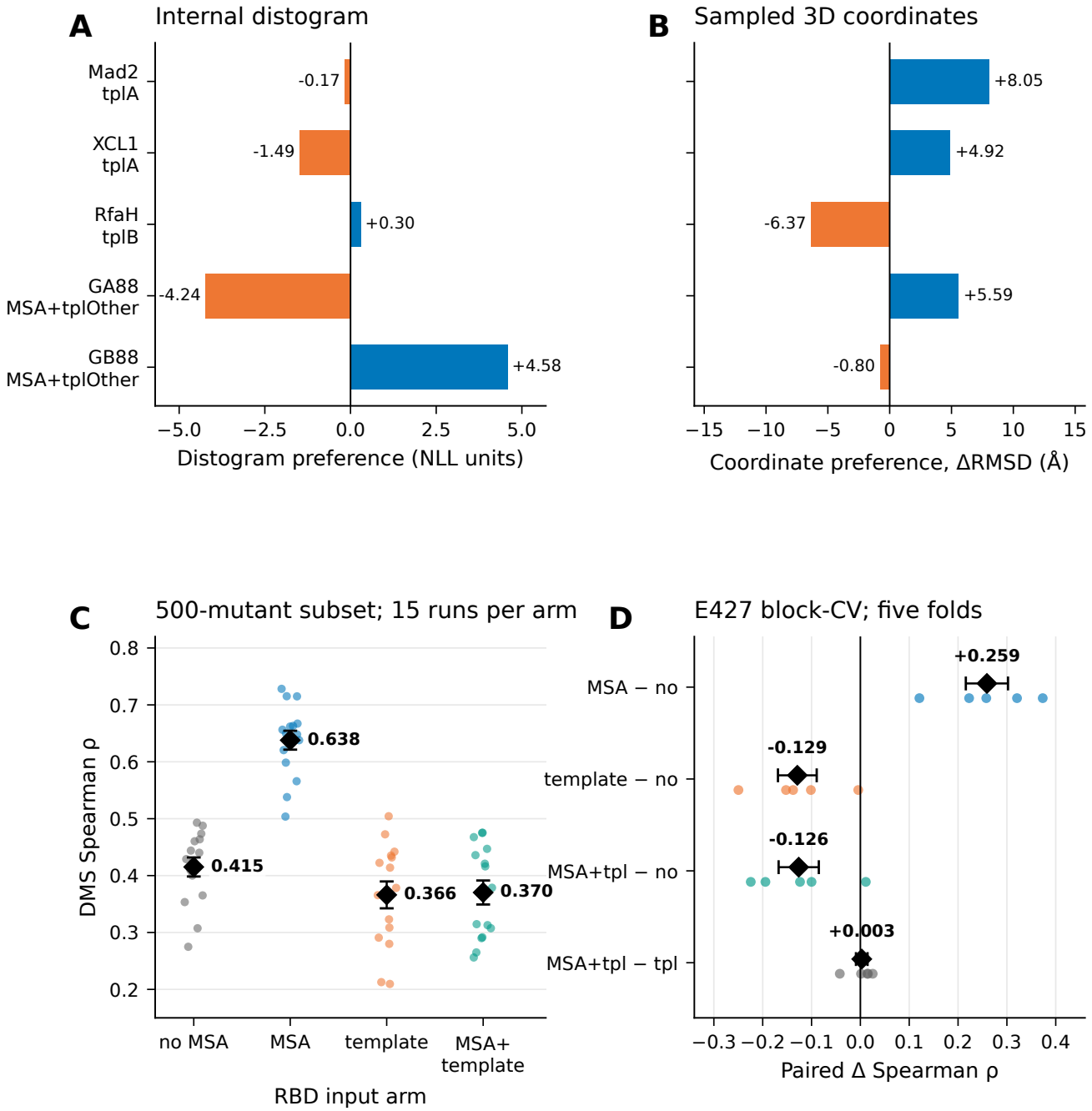


Fig 2. Distogram-coordinate readout decoupling under template conditioning and opposite functional effects of templates vs. MSAs on RBD mutation prediction. (A) Internal trunk distogram preferences (pref = $\mathrm{NLL}_B - \mathrm{NLL}_A$, in NLL units) and (B) explicit 3D sampled coordinate preferences ($\Delta\mathrm{RMSD} = \mathrm{RMSD}_B - \mathrm{RMSD}_A$, in Å, Protocol E428; $n = 16$ structures per cell) across five diagnostic template-conditioned arms (`Mad2 tplA`, `XCL1 tplA`, `RfaH tplB`, `GA88 MSA+tplOther`, and `GB88 MSA+tplOther`), demonstrating directional sign inversions and readout decoupling between trunk distograms and diffusion-sampled 3D coordinates. (C) Cross-position DMS Spearman ρ on the 500-mutant SARS-CoV-2 RBD subset across four input arms (`no MSA`, `MSA`, `template 6M0J`, and `MSA+template`; 15 runs per arm from 5 splits × 3 initializations shown as individual points with black diamonds indicating mean ± SE). (D) Split-paired Δ Spearman ρ contrasts under 5-fold contiguous sequence-block cross-validation (Protocol E427; individual fold deltas shown alongside mean ± SE), confirming that the wild-type crystal template impairs RBD mutation-effect prediction (−0.129) and suppresses the beneficial MSA write in the combined arm.

- **CASP3 homodimer interface amplification without paired coevolution:** Across 392 stride-4 CASP3 mutants, the inter-chain A–B pair block showed higher directional cosine (0.749 vs. 0.712 in A–A/B–B, difference +0.037, paired Wilcoxon $p = 5.5 \times 10^{-66}$) and norm amplification (1.280 vs. 1.054/1.053). Shuffling chain-B MSA rows across 100 mutants to break species-level pairing preserved identical amplification (A–B: 0.749/1.280; A–$B - A$–$A = +0.037$, $p = 3.9 \times 10^{-18}$; paired vs. unpaired embedding cosine 0.969), indicating that MSA context amplifies an existing trunk interface prior without requiring paired cross-chain coevolution.

3 Discussion

3.1 Persistent MSA writing in an AF3-style pair representation

Although AF3-style architectures discard the explicit MSA representation after four blocks per recycle, our recycle-gating results on RF3 show that transient MSA processing leaves a persistent imprint in the pair representation. A single recycle ($K = 1$) installed roughly 75% of the geometric compression and 85% of the DMS predictive gain, and nine subsequent MSA-free recycles erased only about 25% of the geometric shift. Thus, within the standard 10-recycle regime, the pair representation retains the initial alignment write across subsequent trunk iterations.

3.2 Evolutionary content rather than rank compression drives mutation-effect gains

Combining geometric metrics with content perturbations clarifies why compressing the mutant representation ensemble does not erase mutation-effect information. Uninformative alignments (`random`, `rowshuffle`) compressed `eff90` nearly as much as real homologs via the shared-context pathway but produced no DMS gain. Functional improvement required column-level amino acid distributions (`colshuffle`, contributing ∼2/3 of the biological gain on RBD) and inter-column covariation (`full15`, contributing ∼1/3), which reorganized mutant perturbation directions (`dCos`) rather than merely reducing dimensionality. Similarly, distillation from single-sequence trunk features and ESM-2 embeddings recovered about 60% of the DMS gain, leaving roughly 40% of the MSA-specific functional advantage tied to alignment-derived features at the tested capacities.

3.3 Target-dependent state selection, distinct template priors, and readout boundaries

Across the conformational panel and explicit 3D sampling ($N = 480$ structures), MSA-induced shifts concentrated in fold-switching proteins and aligned in direction with inferred superfamily consensus states (pulling KaiB and RfaH away from crystal states, and Mad2, XCL1, GA88, and GB88 toward their canonical or native folds). While this multi-level consistency supports viewing the MSA as a family-consensus writer and selector, we emphasize that these observations do not constitute a closed causal proof that the exact same latent subspace direction simultaneously causes both DMS gains and conformational switches.

Comparing MSAs with structural templates showed that templates act as a distinct prior rather than an equivalent evolutionary input. On RBD, a wild-type crystal template impaired DMS prediction and overrode the beneficial MSA effect when combined. In conformational selection, templates strongly steered 3D structures, but conflict resolution between templates and MSAs was target-dependent: templates dominated opposing MSAs in KaiB, RfaH, and Mad2, whereas native MSAs preserved the native fold in GA88 (16/16) and produced a 50/50 ensemble in GB88 against an opposite-fold template. Methodologically, the sign inversions between distogram NLL and 3D coordinates in `tplA` (Mad2, XCL1) and `msa_tplOther` (GA88, GB88) demonstrate that distogram readouts are not template-artifact-free; conformational audits of AF3-style models should therefore include explicit 3D coordinate generation, particularly in template-conditioned settings.

3.4 Limitations

Several scope boundaries apply to these findings. First, all results come from a single RF3 checkpoint and fixed inference setup and cannot be extrapolated without testing to AF2, AF3, or Boltz [3, 4, 30]. Second, DMS analyses used frozen trunk representations across three proteins (RBD, CASP3, CI) with single-point mutants and one assay label each; full de-biasing was completed on RBD and CASP3 but not on the smaller CI dataset ($N = 351$), RBD showed a flattened-linear readout boundary in E427, and the ∼2/3 conservation vs. ∼1/3 covariation split and ∼8-row saturation are specific to the 15-row RBD alignment. Third, two-state evaluations simplify multi-state thermodynamic landscapes; explicit 3D coordinate verification (E428) covered 6 of the 13 panel entries at 16 structures per cell, several arms had large absolute RMSDs (so claims are restricted to relative state preference ΔRMSD rather than experimental accuracy or Boltzmann populations), and distogram NLL and ΔRMSD remain

distinct, non-interchangeable readouts. Finally, template behavior may depend on specific PDB chain choices, truncation, and featurization details [31].

4 Materials and Methods

4.1 Model architecture, inference configuration, and feature extraction

All experiments used a fixed checkpoint of RoseTTAFold3 (RF3) [5] with 10 recycles, 50 diffusion steps, and a diffusion batch size of 2. Within each recycle, the MSA is processed in four MSA blocks (using outer-product mean operations) to update the residue-pair representation, after which the explicit MSA representation is discarded. We extracted the trunk pair tensor $\mathbf{Z}_{\mathrm{II}} \in \mathbb{R}^{L\times L\times 128}$ immediately after the distogram head projection. For single-point mutants, the true mutated residue index i was sliced along both the row and column axes of $\mathbf{Z}_{\mathrm{II}}$ and stacked into a $2 \times L \times 128$ tensor (for homodimeric CASP3, the $488 \times 488 \times 128$ matrix was archived and sliced at the monomer mutation coordinates; this avoided an early position-0 slicing bug that artificially produced `eff90` ≈ 30). Per-mutant language model representations (`FAESM`, $L \times 1280$) were extracted from the final layer of ESM-2 650M (`esm2_t33_650M_UR50D`) [14].

4.2 DMS datasets and representation geometry metrics

Three DMS datasets were analyzed: (i) SARS-CoV-2 Spike RBD (201 aa, ACE2 binding `bind_avg`, $N = 3,998$ single-point mutants, 15 curated sarbecovirus MSA rows, plus a stride-8 subset of $N = 500$ mutants for E1, E7, and E49-RBD) [7]; (ii) human CASP3 (244 aa/chain homodimer, 488 tokens, fluorescence activity, $N = 1,469$ single-point mutants, 69 Swiss-Prot homolog rows) [22]; and (iii) bacteriophage λ repressor CI (237 aa construct, 59 assayed positions, ProteinGym `DMS_score`, $N = 351$ mutants, 512 UniRef30 rows) [9].

Over the centered mutant feature matrices, we computed: (i) effective rank (`eff90`), the number of principal components explaining 90% of total variance; (ii) linear CKA [29] between input conditions; (iii) directional cosine (`dCos` / Δ direction), the median cosine similarity of per-mutant difference vectors ($\mathbf{z}_{\mathrm{mutant}} - \mathbf{z}_{\mathrm{WT}}$) across conditions; and (iv) Δ norm ratio, the median Euclidean norm ratio of $\mathbf{z}_{\mathrm{mutant}} - \mathbf{z}_{\mathrm{WT}}$ relative to the no-MSA baseline.

4.3 DMS regression, de-biasing, block cross-validation, fusion, and distillation

Spearman ρ values were compared strictly within each evaluation protocol:

- **Canonical full-set protocol:** Frozen features were trained with the `SinglePredictorV2` transformer head, per-position/per-channel z-score standardization, 70%/30% random position holdout (5 splits $\times$ 3 initializations), 300 epochs, AdamW (lr $= 10^{-4}$, weight_decay $= 0.05$), cosine schedule, MSE loss, and best-epoch validation selection.
- **De-biasing protocols:** Dual-validation separated held-out positions into validation (epoch selection) and test (reporting) subsets; strict train-only standardization computed per-channel mean and variance using training positions only.
- **Protocol E427 (contiguous-block CV and lightweight probes):** Sorted mutated positions were partitioned into 5 contiguous blocks (20% positions/fold, 3 initializations). Lightweight readout heads tested a per-residue MLP and a flattened linear probe. E427 also re-evaluated the four RBD template arms using `SinglePredictorV2` and per-channel standardization.
- **Protocols E1 and E7 (500-mutant RBD subset):** Used a lighter per-position head and per-position scalar standardization; E1 referenced biological gains to `depth01`.
- **Three-stream fusion, E429 paired noise floor, and D1 distillation:** Multi-stream fusion ($\mathbf{Z}_{\mathrm{no}}$, $\mathbf{Z}_{\mathrm{msa}}$, `FAESM`) used cross-attention with an orthogonality penalty and dead-stream/zeros controls. Protocol E429 ran 16 identical training replicates across two independent processes (8 runs each) on fixed split and initialization for RBD `x_msa_faesm`. Protocol D1 trained linear, MLP-256/512, and 1–8-layer transformer models (plus one-hot amino-acid controls and CKA-objective separate encoders) to reconstruct $\mathbf{Z}_{\mathrm{msa}}$ from $\mathbf{Z}_{\mathrm{no}}$ and/or `FAESM`.

4.4 Conformational state selection: distogram preference and 3D coordinate sampling

The 13-entry two-state panel (S2 Table) was scored on the RF3 64-bin distogram (2–22 Å plus no-contact bin) over the shared resolved residue universe (`range matching`, avoiding span mismatches and the 14-residue RfaH loop

offset) using reference Cα–Cα distance matrices:

$$\text{pref} = \text{NLL}_B - \text{NLL}_A, \qquad \Delta\text{pref}_{\text{MSA}} = \text{pref}_{\text{MSA}} - \text{pref}_{\text{noMSA}}, \tag{1}$$

where pref > 0 indicates preference for State A (empirical repeat-forward shift noise band ± 0.14 to ± 0.27). Template inputs required non-zero templated-atom counts and preserved original chain identifiers.

In Protocol E428, explicit 3D coordinates were sampled for KaiB, RfaH, Mad2, and XCL1 across {`nomsa, msa, tplA, tplB, msa_tplA`} and for GA88 and GB88 across {`nomsa, msa, tplOwn, tplOther, msa_tplOther`}, with 8 repetitions $\times$ 2 diffusion samples = 16 structures per cell ($N = 480$ sample-level `*sample-*_model.cif` files). Query Cα coordinates were aligned to reference CIFs via Needleman–Wunsch alignment and Kabsch superposition over shared residues to compute $\Delta\text{RMSD} = \text{RMSD}_B - \text{RMSD}_A$ ($\Delta\text{RMSD} > 0$ closer to State A).

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Competing Interests

The author declares that no competing interests exist.

Author Contributions

Cheng Ding: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Data Availability Statement

Analysis code and the numerical outputs underlying all reported summary statistics will be available at https://github.com/Lizon1c/rf3-msa-template-audit. The 480 sampled coordinate files and analysis-ready feature archives are available at https://doi.org/10.5281/zenodo.23758807. Publicly available structural and DMS input datasets are identified in Materials and Methods.

Supporting Information Captions

S1 Fig. Pair-representation effective rank and CKA across three DMS proteins. Effective rank (`eff90`), linear CKA, and directional cosine (`dCos`) across SARS-CoV-2 RBD, human CASP3, and λ repressor CI (archived panel: `fig1_geometry_collapse.png`).

S2 Fig. RBD alignment dose-content decomposition and recycle-gating trajectories. Effective rank (`eff90`) and Spearman ρ across the 9-arm Protocol E1 series and Protocol E7 recycle-gating series on the 500-mutant RBD subset (archived panel: `fig2_arms_effrank.png`).

S1 Text. Extended methods, analysis artifact notes, and supporting information accounting. Detailed description of mutation-site slicing verification, standardization leakage analysis, range-matching alignment fixes, template atom-count checks, Protocol D1 architecture ladder, Protocol E429 paired noise floor, and CASP3 unpaired interface controls.

S1 Table. Complete DMS de-biasing and Protocol E427 contiguous-block cross-validation and lightweight probe results. Per-split and per-fold Spearman ρ values for canonical, dual-validation, strict

train-only standardization, 5-fold contiguous block-CV, per-residue MLP, and flattened linear probes across RBD, CASP3, and CI.

S2 Table. Complete 13-entry two-state distogram preference panel and five-target distogram template steering results. Range-matched residue counts, MSA depths, $\text{pref}_{\text{noMSA}}$, pref_{MSA}, shift values, and distogram-level template arms (`tplA`, `tplB`, `msa_tplA`, `steer`, `conflict`).

S3 Table. Complete 30-cell explicit 3D coordinate sampling benchmark (Protocol E428, $N = 480$ structures). Mean RMSD_A, mean RMSD_B, ΔRMSD, and fraction of structures closer to State A across all 16 structures (8 repetitions $\times$ 2 diffusion samples) for each of the 30 target $\times$ arm cells.

S4 Table. Three-stream fusion marginals, Protocol D1 reconstruction ladder, Protocol E429 paired noise-floor replicates, and CASP3 homodimer interface block statistics. Complete numerical tables for $\mathbf{Z}_{\text{no}}/\mathbf{Z}_{\text{msa}}/\texttt{FAESM}$ fusion, RBD learning-rate ($\lambda = 0.25$) and input-noise interventions, 16-run E429 paired noise series, D1 CKA/R^2/DMS recovery, and paired vs. unpaired CASP3 A–A, B–B, and A–B blocks.

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Supporting Information Tables

S1 Table. Complete DMS de-biasing and Protocol E427 contiguous-block cross-validation and lightweight probe results

Target	Protocol / Probe	$\Delta\rho$ (MSA − no)	Consistency	Fold / Split Breakdown & Notes
RBD	Canonical full-set (`SinglePredictorV2`)	+0.179 / +0.204	5/5 splits	Two independent full-set runs; best-validation selection
RBD	De-biased: test at val-selected epoch	+0.141	5/5 splits	Removes single-validation epoch-selection optimism (+0.04–0.09)
RBD	De-biased: best held-out test	+0.143	5/5 splits	Dual-validation split estimate
RBD	De-biased: strict train-only z-score	+0.146	15/15 runs	MSA-only 0.644 → 0.589 (−0.055); no-MSA −0.022
RBD	E427 Contiguous block-CV (Transformer)	+0.256	5/5 folds	Folds: +0.200, +0.328, +0.312, +0.214, +0.226
RBD	E427 Per-residue nonlinear MLP	+0.157	5/5 splits	Matches canonical transformer probe magnitude
RBD	E427 Flattened linear probe	+0.030	3/5 splits	Readout boundary: both arms $\rho \approx 0.30$
CASP3	Canonical full-set (`SinglePredictorV2`)	+0.162	5/5 splits	Homodimer 488 × 488 × 128 sliced at true mutant positions
CASP3	De-biased: dual-validation range	+0.142 to +0.183	5/5 splits	Across dual-validation estimators
CASP3	E427 Contiguous block-CV (Transformer)	+0.264	5/5 folds	Folds: +0.330, +0.141, +0.267, +0.351, +0.231
CASP3	E427 Per-residue nonlinear MLP	+0.160	5/5 splits	Preserves full transformer probe gain
CASP3	E427 Flattened linear probe	+0.163	5/5 splits	Linearly decodable MSA write
CI	Canonical full-set (`SinglePredictorV2`)	+0.075	4/5 splits	$N = 351$ mutants across 59 positions; not fully de-biased
CI	E427 Per-residue nonlinear MLP	+0.089	5/5 splits	Consistent gain across all splits
CI	E427 Flattened linear probe	+0.085	3/5 splits	Direction preserved with lower split consistency

S2 Table. Complete 13-entry two-state distogram preference panel and five-target distogram template steering results

Target	State A / State B (PDB)	MSA	Res.	no	MSA	Shift	tplA	tplB	MSA+tplA (Steer / Conflict)
KaiB	2QKE XR GS / 5JYT NMR FS	257	90	+0.72	−1.57	−2.29	+3.59	−3.00	+3.36 (−6.59 / −0.23)
RfaH	2OUG XR α / 2LCL NMR β	257	42	+0.10	−2.89	−2.99	+1.49	+0.30	+1.45 (−1.19 / −0.04)
Mad2	1DUJ NMR O / 1S2H NMR C	257	184	−0.05	−0.96	−0.91	−0.17	−1.87	−0.17 (−1.70 / 0.00)
XCL1	1J8I chemokine / 2JP1 β-sand.	160	39	−0.78	+2.09	+2.87	−1.49	−0.86	−1.44 (+0.63 / +0.05)
LacY	1PV6 inward / 4OAA outward	257	388	+0.06	−0.01	−0.08	+0.54	+0.08	+0.51 (−0.47 / −0.03)
selecase	4QHF mon. / 4QHJ olig.	2	105	+0.13	+0.13	−0.01	—	—	Null shift
GA88/GB88	2JWS 3α / 2JWU α/β (main)	24	40	+0.52	+3.07	+2.56	—	—	E49b GA: +0.38 → +2.74; GB: −0.23 → −2.66
β_2AR	2RH1 inact. / 3SN6 act.	257	271	−0.06	+0.09	+0.15	—	—	Null shift
Abl1	2GQG DFG-in / 2HYY DFG-out	257	262	≈ 0	+0.01	+0.01	—	—	Null shift
EGFR	1M17 αC-in / 1XKK αC-out	257	265	+0.14	+0.12	−0.02	—	—	Null shift
ubiquitin	1UBQ XR / 1D3Z NMR	257	76	−0.08	−0.10	−0.03	—	—	Null shift
GB1	1PGB XR / 2GB1 NMR	89	55	+0.36	+0.44	+0.09	—	—	Null shift
thioredoxin	2TRX XR / 1XOA NMR	257	108	+0.26	+0.38	+0.13	—	—	Null shift

S3 Table. Complete 30-cell explicit 3D coordinate sampling benchmark (Protocol E428, $N = 480$ structures)

Target	Arm	RMSD A (Å)	RMSD B (Å)	ΔRMSD (Å)	Closer to A	Coordinate Verification Notes ($n = 16$ structures/cell)
KaiB	`nomsa`	4.737	8.912	+4.174	100.0%	Sequence-only favors crystallized ground state A
KaiB	`msa`	9.524	4.826	−4.698	0.0%	MSA shifts 16/16 structures to fold-switched state B
KaiB	`tplA`	2.836	9.588	+6.752	100.0%	Template A steers 16/16 to state A
KaiB	`tplB`	8.944	3.073	−5.871	0.0%	Template B steers 16/16 to state B
KaiB	`msa_tplA`	3.083	9.432	+6.349	100.0%	Joint conflict arm follows `tplA`
RfaH	`nomsa`	12.296	12.482	+0.185	56.2%	Near 50/50 split without MSA
RfaH	`msa`	15.065	5.274	−9.791	0.0%	MSA strongly steers 16/16 to β-barrel state B
RfaH	`tplA`	0.824	13.702	+12.879	100.0%	Template A closely copies α-hairpin state A
RfaH	`tplB`	13.517	7.148	−6.369	0.0%	Template B steers 16/16 to state B (despite +0.30 distogram)
RfaH	`msa_tplA`	5.786	13.383	+7.598	100.0%	Joint conflict arm follows `tplA`
Mad2	`nomsa`	12.982	13.749	+0.767	87.5%	Modest sequence-only preference for open state A
Mad2	`msa`	9.836	5.694	−4.142	0.0%	MSA steers 16/16 to closed state B
Mad2	`tplA`	3.140	11.187	+8.046	100.0%	Steers 16/16 to open state A (resolving −0.17 distogram)
Mad2	`tplB`	10.656	1.013	−9.643	0.0%	Steers 16/16 to closed state B
Mad2	`msa_tplA`	3.581	11.311	+7.730	100.0%	Joint conflict arm follows `tplA`
XCL1	`nomsa`	13.479	9.478	−4.000	0.0%	Sequence-only favors β-sandwich state B
XCL1	`msa`	6.111	12.862	+6.751	100.0%	MSA steers 16/16 to chemokine state A
XCL1	`tplA`	8.023	12.942	+4.919	100.0%	Steers 16/16 to state A (resolving −1.49 distogram)
XCL1	`tplB`	14.171	10.718	−3.453	6.2%	Steers 15/16 to state B
XCL1	`msa_tplA`	9.632	14.158	+4.526	93.8%	Co-directional in 3D coordinates (15/16 at state A)
GA88	`nomsa`	11.400	9.221	−2.179	6.2%	Sequence alone does not stably select GA fold (1/16 A)
GA88	`msa`	5.255	11.937	+6.682	100.0%	Native MSA restores 3α fold A (16/16)
GA88	`tplOwn`	4.875	11.798	+6.923	100.0%	Own template selects 3α fold A (16/16)
GA88	`tplOther`	13.834	1.455	−12.378	0.0%	Opposite template alone flips 16/16 to GB fold B
GA88	`msa_tplOther`	6.008	11.601	+5.593	100.0%	Native MSA preserves GA fold A (16/16; distogram −4.24)
GB88	`nomsa`	13.405	10.348	−3.058	0.0%	Sequence alone selects native GB fold B (16/16)
GB88	`msa`	13.605	1.218	−12.387	0.0%	Native MSA sharpens α/β fold B (16/16)
GB88	`tplOwn`	13.597	0.837	−12.761	0.0%	Own template selects α/β fold B (16/16)
GB88	`tplOther`	3.759	12.630	+8.871	100.0%	Opposite template alone flips 16/16 to GA fold A
GB88	`msa_tplOther`	11.586	10.790	−0.796	50.0%	Native MSA yields 50/50 mixed ensemble (distogram +4.58)

S4 Table. Three-stream fusion marginals, D1 distillation, E429 paired noise floor, and CASP3 interface blocks

Audit Module	Condition / Metric	Archived Numerical Result & Interpretation
Three-stream fusion	RBD ($\mathbf{Z}_{\text{no}}, \mathbf{Z}_{\text{msa}}$, `FAESM`)	Single-stream ρ: 0.465, 0.644, 0.484; `FAESM` marginal on $\mathbf{Z}_{\text{msa}}$: -0.005 (vs. $+0.075$ on $\mathbf{Z}_{\text{no}}$); $\mathbf{Z}_{\text{no}}$ marginal on $\mathbf{Z}_{\text{msa}}$ + `FAESM`: $+0.002$.
Three-stream fusion	CASP3 ($\mathbf{Z}_{\text{no}}, \mathbf{Z}_{\text{msa}}$, `FAESM`)	Single-stream ρ: 0.394, 0.556, 0.584; `FAESM` marginal on $\mathbf{Z}_{\text{msa}}$: $+0.051$ (5/5 splits); $\mathbf{Z}_{\text{no}}$ marginal on $\mathbf{Z}_{\text{msa}}$ + `FAESM`: -0.001.
Three-stream fusion	CI ($\mathbf{Z}_{\text{no}}, \mathbf{Z}_{\text{msa}}$, `FAESM`)	Single-stream ρ: 0.374, 0.449, 0.452; `FAESM` marginal on $\mathbf{Z}_{\text{msa}}$: -0.039 (unresolved, driven by one outlier split); $\mathbf{Z}_{\text{no}}$ marginal: $+0.008$.
Gradient	RBD $\lambda = 0.25$ LR scaling	Late-training gradient ratios $f/z \approx 6.5$, $f/m \approx 2.7$ (CASP3/CI $f/m \approx 3.5$–3.8). Scaling `FAESM` encoder LR by $\lambda = 0.25$ gives split deltas $+0.0155, +0.0252, +0.0091, +0.0146, +0.0069$; mean $+0.0143 \pm 0.0032$ paired SE ($t = 4.48$, 5/5).
Gradient	RBD `FAESM` input noise	75% Gaussian noise shifts ρ from 0.649 $\rightarrow$ 0.662 ($+0.0128 \pm 0.0110$ SE, 3/5 splits); 100% noise returns to 0.648 (near dead-stream control 0.655).
Protocol E429	Paired training noise floor	16 identical runs (2 processes $\times$ 8 runs) on RBD `x_msa_faesm`: within-process SD 0.0016 and 0.0027; 15/16 runs in 0.6683–0.6732 (full range 0.012 with one outlier at 0.6608).
Protocol D1	RBD $\mathbf{Z}_{\text{msa}}$ distillation	Validation CKA: linear 0.74, MLP-512 and 1–8-layer transformers 0.80–0.83 (coordinate R^2 : 0.25 $\rightarrow$ 0.36; 8-layer train–val CKA gap $+0.09$; one-hot AA control CKA ≈ 0.50). Separate CKA encoders (depths 2/4/6/8): both 0.811 > `FAESM`-only 0.75–0.76 > $\mathbf{Z}_{\text{no}}$-only 0.716–0.717. Functional DMS ρ: $\mathbf{Z}_{\text{no}}$ 0.446, reconstructed 0.575, real $\mathbf{Z}_{\text{msa}}$ 0.663 (59.4% recovery; $\sim$40% unrecovered).
CASP3 interface	Paired vs. unpaired MSA	Paired (392 mutants): *A*–*A* cos 0.712 / norm 1.054; *B*–*B* 0.712/1.053; *A*–*B* 0.749/1.280 (*A*–*B* − *A*–*A* = $+0.037$, $p = 5.5 \times 10^{-66}$). Chain-B shuffled unpaired (100 mutants): *A*–*A* 0.713/1.055; *B*–*B* 0.712/1.054; *A*–*B* 0.749/1.280 ($+0.037$, $p = 3.9 \times 10^{-18}$); paired vs. unpaired embedding cosine 0.969.